

DAX for creating tables and columns

DAX FUNCTIONS IN POWER BI



Carl Rosseel

Curriculum Manager at DataCamp

DAX stands for data analysis expressions

- DAX is a formula expression language used in multiple Microsoft analytics tools



- DAX formulas include functions, operators and values to perform advanced calculations
- DAX formulas are used in:
 - **Measures**
 - **Calculated columns**
 - **Calculated tables**
 - Row-level security

The power of DAX

- It opens up new capabilities:
 - Joins, filters, measures, and calculated fields become part of your toolbox
- DAX + Power Query = a powerful data analysis tool:
 - Dive deeper into the data and extract key insights
 - Use DAX for rapid prototyping

Measures vs calculated columns

Calculated Columns:

- Calculated on data import
- Visible in Table and Report view

```
COST = Orders[Sales] - Orders[Profit]
```

Order_ID	Sales	Profit	Cost
3151	\$77.88	\$3.89	\$73.99
3152	\$6.63	\$1.79	\$4.84
3153	\$22.72	\$10.22	\$12.50
3154	\$45.36	\$21.77	\$23.59

Measures vs calculated columns

Calculated Columns:

- Calculated on data import
- Visible in Table and Report view

```
COST = Orders[Sales] - Orders[Profit]
```

Order_ID	Sales	Profit	Cost
3151	\$77.88	\$3.89	\$73.99
3152	\$6.63	\$1.79	\$4.84
3153	\$22.72	\$10.22	\$12.50
3154	\$45.36	\$21.77	\$23.59

Measures:

- Calculated at query run-time
- Visible only in report pane

```
Total Sales = SUM(Orders[Sales])
```

Region	Total Sales
Central	\$501,239.89
East	\$678,781.24
West	\$391,721.91
South	\$725.457.82
Total	\$2,297,200.86

Context allows you to perform dynamic analysis

There are three types of context: row, query, and filter context

- Row context: (1)
 - "The current row"
 - DAX calculated columns

```
COST = Orders[Sales] - Orders[Profit]
```

Context allows you to perform dynamic analysis

There are three types of context: row, query, and filter context

- Row context: (1)
 - "The current row"
 - DAX calculated columns

```
COST = Orders[Sales] - Orders[Profit]
```

Order_ID	Sales	Pofit	Cost
3151	\$77.88	\$3.89	\$73.99
3152	\$6.63	\$1.79	\$4.84
3153	\$22.72	\$10.22	\$12.50
3154	45.36	\$21.77	\$23.59

Context allows you to perform dynamic analysis

There are three types of context: row, query, and filter context

- Query context: (2)
 - Refers to the subset of data that is implicitly retrieved for a formula
 - Controlled by slicers, page filters, table columns and row headers
 - Controlled by chart/visual filters
 - Applies after row context

Context allows you to perform dynamic analysis

- Query context: (2)
 - *Example:* Filter data by Region.
- Query context: (2)
 - *Example:* Filter data by State.

Region	Total Sales
Central	\$501,239
East	\$678,781
West	\$391,721
South	\$725.457

State	Total Sales
Alabama	\$13,724
Arizona	\$38,710
Arkansas	\$7,669
California	\$381,306

Context allows you to perform dynamic analysis

There are three types of context: row, query, and filter context

- Filter Context: (3)
 - The set of values allowed in each column, or in the values retrieved from a related table
 - By using arguments to a formula or by using report filters on row and column headings
 - Applies after query context

Context allows you to perform dynamic analysis

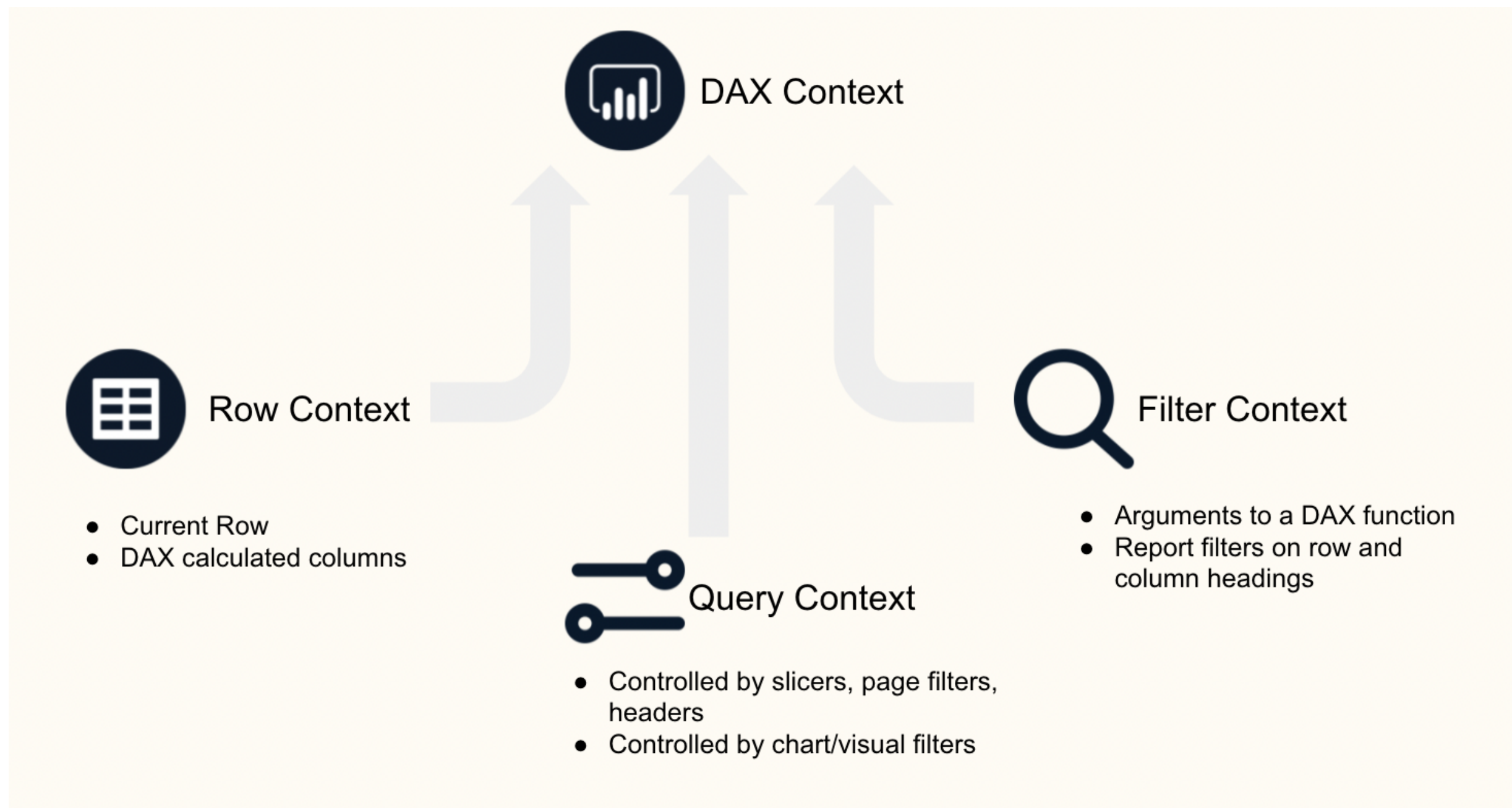
There are three types of context: row, query, and filter context.

- Filter Context (3)

```
Total Costs East = CALCULATE([Total Costs], Orders[Region] = 'East')
```

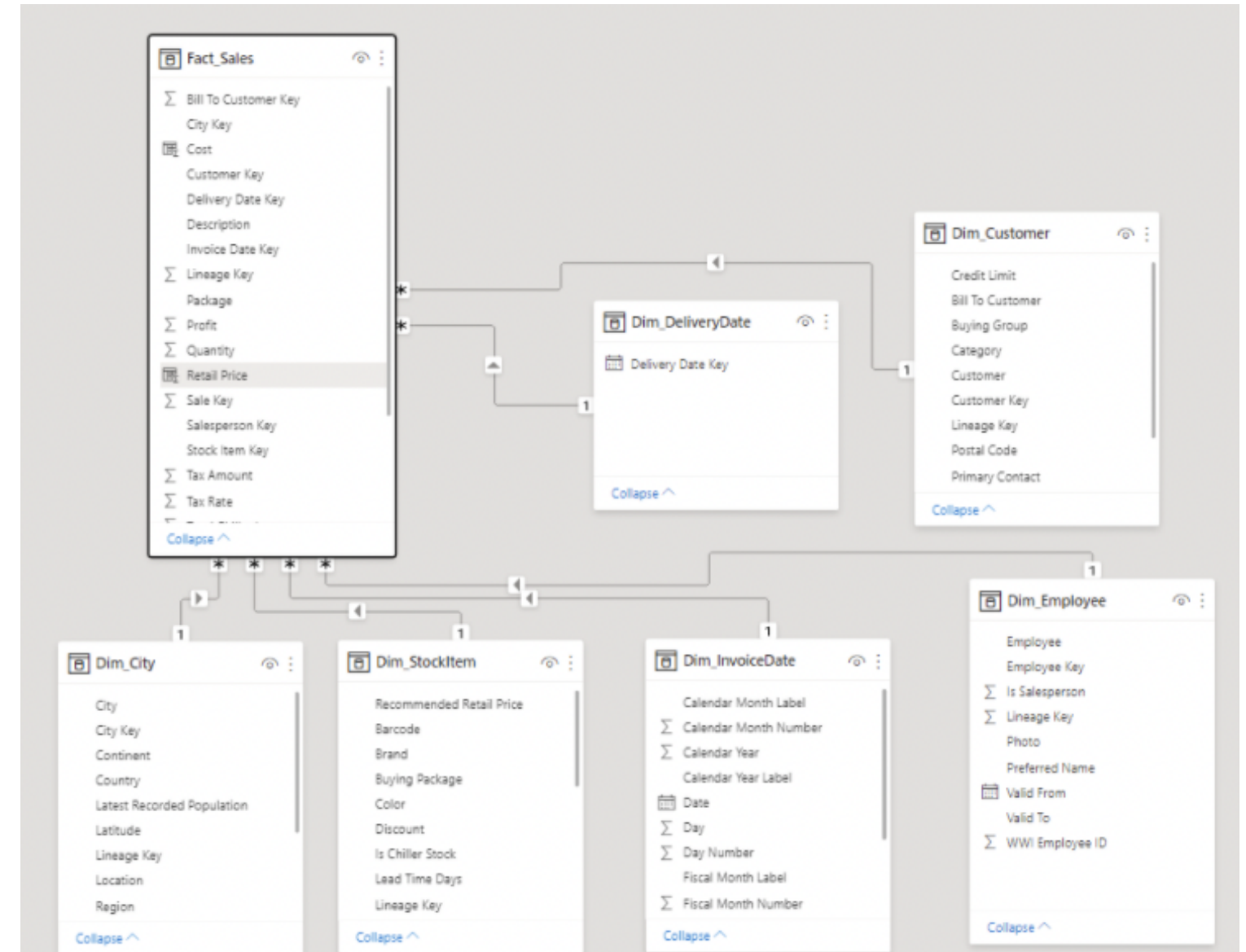
Region	Total costs	Total costs East
Central	\$617,039	
East	\$587,258	\$587,258
West	\$461,534	
South	\$344,972	
Total	\$2,010,804	\$587,258

Context in a nutshell



World wide importers dataset

- A fictitious wholesaler who imports and distributes novelty goods
- The dataset consists of:
 - A fact table that detailing sales transactions
 - Multiple other dimension tables:
 - Dates
 - Customers
 - Cities
 - Employees
 - Stock Items



Let's practice!

DAX FUNCTIONS IN POWER BI

DAX for calculated tables and columns

DAX FUNCTIONS IN POWER BI



Carl Rosseel
Curriculum Manager

Let's practice!

DAX FUNCTIONS IN POWER BI